

## N'GUIGMI, Niger

WMO: 610490

Lat: 14.250N

Lon: 13.117E

Elev: 286

StdP: 97.94

Time Zone: 1.00 (E01)

Period: 90-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 11.6       | 13.0 | -7.9                          | 2.0 | 28.7 | -7.0 | 2.2 | 27.0 | 8.9                   | 21.7 | 8.0 | 23.5 | 0.2                   | 40   | 0.378 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 15.2                   | 43.6            | 21.3 | 42.7 | 21.3 | 41.7 | 21.3 | 26.4                | 34.4 | 25.9 | 34.0 | 25.5 | 33.6 | 1.9                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.8                            | 20.5 | 28.6 | 24.2 | 19.8 | 28.4 | 23.7 | 19.2 | 28.4 | 83.9          | 33.7 | 81.8 | 34.0 | 80.1 | 33.5 | 31.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 7.0               | 5.7  | 4.5 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |    |    |
|                   |      |     | 3.5                        | 28.8 | 0.6                | 1.4 | 3.1                                                | 29.8 | 2.7        | 30.7 | 2.4        | 31.5 | 2.0        | 32.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 29.1   | 21.5  | 24.6  | 28.5  | 33.0  | 34.4  | 34.1  | 31.8  | 29.9  | 31.4  | 30.9  | 26.6  | 22.3 |
|                                                                                  | DBStd       | 4.83   | 2.88  | 3.37  | 2.83  | 2.12  | 1.65  | 1.83  | 2.17  | 1.92  | 1.64  | 1.59  | 2.40  | 2.70 |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 6      | 4     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    |
|                                                                                  | CDD10.0     | 6969   | 356   | 409   | 573   | 690   | 755   | 723   | 676   | 617   | 641   | 649   | 497   | 383  |
|                                                                                  | CDD18.3     | 3933   | 102   | 177   | 314   | 440   | 497   | 473   | 418   | 359   | 391   | 391   | 247   | 126  |
|                                                                                  | CDH23.3     | 55003  | 1171  | 2137  | 4298  | 6817  | 7888  | 7332  | 5858  | 4511  | 5292  | 5418  | 2923  | 1359 |
|                                                                                  | CDH26.7     | 33995  | 497   | 1128  | 2621  | 4682  | 5516  | 4986  | 3512  | 2345  | 3141  | 3382  | 1607  | 578  |
| Wind                                                                             | WSAvg       | 1.2    | 1.5   | 1.5   | 1.6   | 1.3   | 1.0   | 1.2   | 1.3   | 0.9   | 0.8   | 1.1   | 1.3   | 1.4  |
| Precipitation                                                                    | PrecAvg     | 192    | 0     | 0     | 0     | 1     | 4     | 15    | 57    | 102   | 28    | 2     | 0     | 0    |
|                                                                                  | PrecMax     | 492    | 0     | 0     | 5     | 20    | 20    | 75    | 173   | 308   | 107   | 45    | 0     | 0    |
|                                                                                  | PrecMin     | 26     | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 4     | 0     | 0     | 0     | 0    |
|                                                                                  | PrecStd     | 100    | 0     | 0     | 1     | 3     | 6     | 16    | 39    | 57    | 27    | 7     | 0     | 0    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 36.3  | 40.1  | 42.2  | 44.7  | 45.0  | 43.8  | 41.8  | 39.4  | 41.2  | 41.0  | 38.8  | 36.2 |
|                                                                                  |             | MCWB   | 17.0  | 18.3  | 19.5  | 21.1  | 21.4  | 22.0  | 23.4  | 24.9  | 21.8  | 21.5  | 20.0  | 17.8 |
|                                                                                  | 2%          | DB     | 33.9  | 37.8  | 40.4  | 43.4  | 43.8  | 42.7  | 40.0  | 37.3  | 39.9  | 40.2  | 37.3  | 33.8 |
|                                                                                  |             | MCWB   | 15.7  | 17.4  | 18.6  | 20.8  | 21.2  | 22.6  | 24.1  | 25.1  | 22.2  | 21.0  | 18.8  | 16.0 |
|                                                                                  | 5%          | DB     | 31.7  | 35.6  | 38.8  | 42.4  | 42.9  | 41.6  | 38.6  | 36.1  | 38.9  | 39.3  | 35.8  | 32.0 |
|                                                                                  |             | MCWB   | 14.8  | 16.6  | 18.1  | 20.3  | 21.1  | 22.9  | 24.3  | 25.1  | 23.0  | 20.7  | 17.9  | 15.2 |
|                                                                                  | 10%         | DB     | 29.4  | 33.1  | 37.1  | 41.1  | 41.7  | 40.4  | 37.2  | 34.9  | 37.6  | 38.1  | 34.2  | 30.2 |
|                                                                                  |             | MCWB   | 14.0  | 15.7  | 17.2  | 19.9  | 21.2  | 23.2  | 24.3  | 24.9  | 23.5  | 20.6  | 17.2  | 14.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 17.9  | 20.0  | 22.2  | 24.8  | 26.0  | 26.7  | 26.6  | 27.0  | 26.9  | 26.3  | 24.3  | 18.5 |
|                                                                                  |             | MCDB   | 34.7  | 36.4  | 38.3  | 40.0  | 37.4  | 37.1  | 34.9  | 33.5  | 35.2  | 35.4  | 35.1  | 35.7 |
|                                                                                  | 2%          | WB     | 16.3  | 18.4  | 20.2  | 23.3  | 25.0  | 25.7  | 25.8  | 26.3  | 26.0  | 25.0  | 21.0  | 16.8 |
|                                                                                  |             | MCDB   | 32.4  | 35.3  | 36.6  | 38.5  | 37.9  | 36.8  | 34.4  | 32.9  | 34.1  | 33.9  | 34.0  | 32.3 |
|                                                                                  | 5%          | WB     | 15.2  | 17.1  | 18.9  | 22.2  | 24.4  | 25.2  | 25.4  | 25.9  | 25.5  | 24.2  | 18.9  | 15.8 |
|                                                                                  |             | MCDB   | 30.8  | 34.3  | 36.3  | 38.0  | 37.1  | 36.4  | 33.9  | 32.5  | 33.3  | 32.8  | 33.6  | 30.7 |
|                                                                                  | 10%         | WB     | 14.1  | 16.0  | 17.8  | 21.2  | 23.7  | 24.7  | 25.1  | 25.5  | 25.1  | 23.3  | 17.7  | 14.9 |
|                                                                                  |             | MCDB   | 28.8  | 32.5  | 35.9  | 37.4  | 36.3  | 35.8  | 33.4  | 31.9  | 32.6  | 32.5  | 32.7  | 28.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 16.0  | 16.3  | 16.3  | 16.3  | 15.2  | 13.1  | 10.6  | 9.7   | 12.6  | 15.0  | 15.5  | 15.4 |
|                                                                                  |             | MCDBR  | 18.6  | 18.8  | 18.4  | 17.5  | 17.0  | 14.6  | 12.2  | 11.3  | 14.4  | 16.0  | 16.5  | 17.0 |
|                                                                                  |             | MCWBR  | 8.2   | 7.7   | 7.3   | 6.6   | 6.3   | 5.0   | 3.6   | 2.7   | 4.3   | 5.6   | 6.8   | 7.1  |
|                                                                                  | 5% WB       | MCDBR  | 18.3  | 17.9  | 17.5  | 15.5  | 13.4  | 12.3  | 10.6  | 10.0  | 12.0  | 14.3  | 16.1  | 16.5 |
|                                                                                  |             | MCWBR  | 8.5   | 8.2   | 7.6   | 7.0   | 5.2   | 4.3   | 3.3   | 3.0   | 3.8   | 5.7   | 7.8   | 7.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.596  | 0.650 | 0.696 | 0.741 | 0.732 | 0.727 | 0.707 | 0.659 | 0.659 | 0.681 | 0.597 | 0.587 |      |
|                                                                                  | taud        | 1.603  | 1.497 | 1.431 | 1.384 | 1.414 | 1.472 | 1.563 | 1.647 | 1.622 | 1.517 | 1.589 | 1.614 |      |
|                                                                                  | Ebn at Noon | 704    | 689   | 675   | 645   | 640   | 635   | 651   | 691   | 691   | 658   | 697   | 695   |      |
|                                                                                  | Edh at Noon | 257    | 297   | 324   | 339   | 324   | 302   | 276   | 257   | 264   | 287   | 258   | 248   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 5.70   | 6.20  | 6.68  | 6.94  | 6.93  | 6.65  | 6.22  | 6.05  | 6.41  | 6.29  | 5.89  | 5.51  |      |
|                                                                                  | RadStd      | 0.16   | 0.14  | 0.29  | 0.24  | 0.28  | 0.27  | 0.19  | 0.26  | 0.15  | 0.17  | 0.14  | 0.14  |      |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (2 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page